Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/01/2019
PI Name:	Lian, Yongsheng Ph.D.		
Project Title:	Development and Verification of a 3D Nucleate Pool Boiling Simulation Model Using PSI Data		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-B: Use of the NASA Physical Sciences Informatics System – Appendix B
Start Date:	08/15/2016	**End Date:**	08/14/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 8/14/2019 per NSSC information (Ed., 9/19/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sussman, Mark Ph.D. (Florida State University)		
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Task Description:	Nucleate pool boiling has widespread applications in industry due to its efficient high heat transfer rates. It has also been used in space vessels for mission critical thermal management and life support fluids and power conversion systems. Under microgravity, the nucleate boiling phenomena and the heat transfer characteristics are significantly different from those under terrestrial gravity. To further understand the effect of gravity on nucleate boiling, NASA recently conducted nucleate pool boiling experiments on the International Space Station (ISS). Both vapor bubble dynamics and heat transfer characteristics were investigated. The experimental data has been included in the Physical Sciences Informatics (PSI) system. The data can be utilized to develop new analyses and scientific insights in nucleate pool boiling. We propose to utilize PSI data to validate and enhance a new competitive numerical model for further investigation of nucleate pool boiling problems under microgravity conditions. Compared to other models, our model has the following advantages which can lead to a more accurate prediction: (1) our method can better resolve thermal boundary layers (away from triple points) using adaptive mesh refinement, (2) a novel Moment of Fluid multiphase flow algorithm enables us to straightforwardly prescribe a microlayer model in 3D general geometries, (3) the moment of fluid interface capturing algorithm enables us to better handle 3D problems such as bubble merging, (3) our method can better conserve the boiling mass flow rate by directly exploiting the moment-of-fluid (sharp) interface reconstruction for approximating the interface heat flux in the vicinity of multiple (greater than 2) materials. The verified model can be used to further study scenarios not investigated on ISS such as vapor bubble departure and collapse. It can also be used to identify optimal heated surface configuration for better heat transfer characteristics. In addition, the model can be used to study other problems under microgravity such as packed bed reactors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of nucleate pool boiling simulation tool in microgravity based on the Navier-Stokes equations from this proposed work can be used to study other boiling problems in Earth gravity such as engine cooling and cooling in electronic devices. The developed tool is based on a novel interface capturing method (moment of fluid method), which is a generalization of the popular volume of fluid method but the new method considers the centroid information and gives better representation of the interface.
Task Progress:	Progress has been made in both numerical method development and the simulation of nucleate pool boiling simulation under both Earth gravity and microgravity. 1: Numerical method development. One key missing piece in the numerical modeling of nucleate pool boiling is the micro layer. It has been observed in experiments that during nucleate boiling, a thin layer of liquid (thickness of few microns), gets trapped under the vapor dome. It has been shown that when this microlayer of liquid gets trapped, its evaporation increases the bubble growth rate. Microlayer formation depends upon the density ratio, the Jakob number, and the apparent contact angle between the interface and the solid wall. In the contact line regime, such as the simulations of water in Earth gravity and reduced gravity the calculation of microlayer was not required due to the moderate Jakob number, and density ratio parameters. However, in case of Perfluoro-n-hexane as the test liquid, this assumption is no longer valid, and the microlayer heat transfer is required to predict the correct growth rate as well as the heat flux. Typically, after the formation, the microlayer grows, and after reaching a maximum size it starts to deplete or reduce in size, and is completely evaporated, the depletion typically takes place during the bubble departure phase. However, under microgravity conditions the bubble departure is delayed due to the reduction of buoyancy force. Experiments on the International Space Station using perfluoro-n-hexane has shown that the bubble kept on growing to the point of a single large vapor bubble taking up the entire space of the container. No bubble departure was observed during multiple experiments on the ISS. Since the bubble does not enter the departure phase, it means the microlayer depletion has not started in the bubble. For the numerical implementation of this scenario, we use a constant thickness microlayer model to capture the growth of vapor bubble. A microlayer model has been developed and implemented in our multiphase flow solver. There are 3 adjustable parameters in the model (layer width, layer height, and layer angle). We systematically adjust these parameters to match the ISS experiment. It is not clear if these parameters can be used for other materials. 2: Earth gravity simulation An adaptive Moment of Fluid (MoF) method has been used to perform numerical simulations of nucleate boiling of water vapor bubble in a pool of test liquid (water) under different heat transfer conditions. Comparison of results of single bubble dynamics especially bubble growth rate, departure radius, and departure time has been made with different results. Validation with theoretical solution for bubble growth in super-heated liquid under absence of wall heat transfer is completed. Details of this case have also been compared with the numerical model. The MoF results are in good agreement with the previous published theoretical and numerical results. Bubble growth is simulated with an addition of heat transfer and results for this simulation have been compared to both experimental as well as numerical solutions. The MoF simulations are in a better agreement with the experimental solution than the numerical ones, validating the MoF model results to be accurate. 3: Reduced gravity simulation Bubble growth in super-heated liquid with wall-heat transfer is performed under reduced gravity conditions and compared with experimental results. The agreement with the parabolic flight experiment results is good with and without the microlayer model. This is consistent with previous similar study. A dimensional analysis study was conducted to identify the effects of gravity on bubble growth rate and departure time. To further validate the proposed formula, we conducted simulation at other gravity levels and found the proposed formula can accurately predict bubble growth rate and departure time when the fluid is water. 4: Microgravity simulation Simulations are performed for microgravity conditions with perfluoro-n-hexane as the test liquid. Results are in good agreement with the experiments performed on the ISS. The simulation used the microlayer model (without microlayer the bubble growth rate and heat flux cannot match the ISS results). To match the experimental results we systematically adjust the three parameters in the microlayer model. However, more study is needed to find out if these parameters can

	be used for other working fluids.
Bibliography Type:	Description: (Last Updated: 08/06/2025)
Abstracts for Journals and Proceedings	Liu Y, Sussman M, Lian Y, Hussaini MY. "A moment-of-fluid method for diffusion equations on irregular domains in multi-material systems." 71st Annual Meeting of the APS Division of Fluid Dynamics, Atlanta, GA, November 18–20, 2018. Bulletin of the American Physical Society. 2018 Nov;63(13):Abstract ID: BAPS.2018.DFD.M10.2. http://meetings.aps.org/link/BAPS.2018.DFD.M10.2 , Nov-2018
Articles in Other Journals or Periodicals	Banerjee S, Lian Y, Liu Y, Sussman M. "Nucleate Boiling simulations at Earth gravity, reduced gravity & microgravity conditions using an adaptive Moment of Fluid method." Computers and Fluids. Under review as of July 2019. , Jul-2019
Articles in Peer-reviewed Journals	Liu Y, Sussman M, Lian Y. "A moment-of-fluid method for diffusion equations on irregular domains in multi-material systems." Journal of Computational Physics. 2020 Feb 1;402:109017. https://doi.org/10.1016/j.jcp.2019.109017 [NOTE: reported in July 2019 as "Under review as of July 2019."] , Feb-2020
Articles in Peer-reviewed Journals	Liu Y, Sussman M, Lian Y, Hussaini MY, Vahab M, Shoele K. "A novel supermesh method for computing solutions to the multi-material Stefan problem with complex deforming interfaces and microstructure." Journal of Scientific Computing. 2022 Apr;91(1):19. https://doi.org/10.1007/s10915-022-01783-1 , Apr-2022